H2700 Masterclass

Making the most of the PlayStation

(with a little help from the PA)

Overview

- Basic H2700 Use
- CPU optimisations
- GPU optimisations
- Debugging using the PA

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Basic H2700 use

- The H2700 consists of 2 parts
  - A standard playstation devkit
  - A logic analyser (Performance Analyser)
- The PA part helps you to fine tune your games

Using the PA

- The PA monitors 3 buses
  - Main Bus
  - Sub Bus
  - Video Bus
- It only knows about bus activity

Using the PA

- The PA operates independantly of the devkit
- A snapshot of the bus activity is taken
  - The PA can hold approx 7 Vsyncs of data
- This data is then analysed by the PA software
  - PA software interprets raw data to show PS structures

Using the PA

- Make a scan
  - Use trigger function to isolate problems
- Identify bottlenecks.
  - CPU or GPU speed problems
- Change code
- Repeat ad nauseum

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CPU optimisation

- I-cache optimisation
- Data optimisation

I-cache optimisation

- Avoid cache thrashing between routines
- Reduce size of core loops

Data optimisation

- Main memory is slow
- Use scratchpad ram for locals
  - Using stack on scratchpad is good
- Always try to group memory accesses into longs
  - Quicker to pack / unpack in registers

Slow code example

- loop:
-         lb v0,(a0)
-         add a0,a0,1
-         sb v0,(a1)
-         add a1,a1,1
-         bne a0,a2,loop
-         nop

Optimised code example

- lw t0,(a0)
- lw t1,4(a0)
- lw t2,8(a0)
- lw t3,12(a0
- sw t0,(a1)
- sw t1,4(a1)
- sw t2,8(a1)
- sw t3,12(a1)

Main bus reads

- R3300 has load scheduling
  - CPU doesn’t stall while waiting for bus.
- Only 1 read can be pending
- You can only interleave 5 instructions
  - 2 cycles are eaten

Bus writes

- PS has 4 write buffers
  - Up to 4 writes can occur before bus activity
- Writes to same page are quicker
- You can group more than 4 writes in one burst

Scratchpad

- The scratchpad isn't on the bus
- However it is in memory
  - Scratchpad load will stall any uncompleted read

GTE

- GTE memory accesses are same as R3300
- GTE register moves can be interleaved

DMA

- GPU dma is fast (Single cycle)
- SPU dma is slower
- CD dma is very slow
- All dma operations reduce bus time available to CPU
- Bus is allocated on a round robin basis.

Common GPU optimisations.

- Texture cache optimisations
- CLUT thrashing
- Primitive combinations
- Clipping

Debugging using the PA

- GPU Timeouts
- Catching strange problems
- Reflashing the H2700 rom

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Summary

- CPU efficiency is becoming more important for PS games
- In most cases CPU time is greater than GPU time
- Speed can be increased by working with rather than against the PS design

Questions??

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